

JOSE OSA

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EDUCATION

University of Michigan, Ann Arbor, MI

- Aerospace and Aeronautics Engineering Graduate student

Columbia University, New York City, NY

- Bachelor of Science in Materials Science Engineering and Aerospace Engineering Minor, Conferred May 2026. GPA: 3.55, Dean's List Award.

The University of the South, Sewanee, TN

- Bachelor of Arts/Science Physics and Mathematics Minor, Conferred May 2024. GPA: 3.87
- Honors/Awards: Global Honors Award
- President of the Sigma Pi Sigma Physics Society and Member of the Order of the Gown.

Space Architecture Studio, Jeffrey Montes, Online

- Semester studio course taught by Blue Origin's Space Architect Jeffrey Montes, where I architected a Dual Person Spacecraft and presented to industry professionals, Conferred Fall 2025

TECHNICAL SKILLS

Python, Data analysis, Data structures, Mathematica, LaTeX, Excel, Arduino, Adobe Photoshop, Illustrator, Premiere Pro, Gamess, ECM, EDM, Laser Machining, Additive Manufacturing, SEM, Crystallography, Ceramics and Polymer Composites, Metallurgy, Organization skills, Resource management skills, Construction skills, Welding and soldering, Customer Service, Circuit Design, Nanofabrication, Photoresins, 2PP Printing, Mechanical Properties Testing, NanoScribe, Molecular Modeling, Burnman Code, Aerodynamics, Aerospace, Grain Structure, Laser Nucleation, Ultra Thin Metal Films

PROFESSIONAL EXPERIENCE

3D Printing and Modeling of Micro-Combustion and Micro-Rocket Systems, Zoetic Robotics Laboratory, Dr. Aubin, Fall 2026 to Present

- Aerospace engineering research intern with the modeling, printing, and testing of small scale resin-based 3D-printed combustion and rocket systems.

Mineral Physics and Engineering Intern, Dr. Wentzovitch Group, Full-time, Lamont-Doherty Earth Observatory, Summer 2025

- Engineering researcher for the Dr. R. Wentzovitch Minerals project group at the Lamont-Doherty Earth Observatory in collaboration with Columbia University's Geodynamics and Seismology department.

Research Assistant, Kysar Group, Columbia University, Internship New York City; Spring and Fall 2025

- Research assistant for the nanofabrication and testing of medical devices at the nano and micro scale.
- Researcher at the Advanced Science Research Center (ASRC) in the City College of New York.

Columbia University CSA Peer Tutor, Part-time, New York City, NY; Academic year 2024-2025,

- Peer tutor for undergraduate students in Mathematics Calculus Courses

Engineering Research Assistant for DolaGon, Internship, Remote, fall 2022-summer-2023

- Main research assistant in the vehicle engineering project for Terrain and trial locations team.

Science Coordinator and counselor, Skylemar, Full-time, Naples, MN; summer 2022

- Science coordinator for the entire camp activities.

ACADEMIA EXPERIENCE

L3Harris National Security Guardian project

- Development of a GEO and HEO satellite architecture for countering and safeguarding potential spying operations by foreign satellite systems.

Columbia Space Initiative

- **Blue Skies:** NASA Engineering competition group from CSI, which aims to tackle agriculture problems via aeronautical vehicles.
- **RASC-AL:** NASA Engineering competition group from CSI, which aims to design from scratch an entire self-sufficient habitat Mars mission for consideration by future NASA operations in Mars.

Argon Plasma Physics Seminar

- Argon Plasma analysis, Plasma element identification, Plasma data acquisition and analysis

3D Harmonic Oscillator in Quantum Mechanics with Applications to Group 6 Hexafluorides

- Experimented with computational analysis using Mathematica to model three dimensional quantum atoms.
- Pending Paper publishing.

Nucleation Growth Stage Characterization of Ultra-Thin Metal Films via Laser Shocking and AFM Imaging: A Mechanistic Framework with Implications for Aerospace Materials Processing

- Laser-induced nucleation and growth phenomena in ultra-thin metallic films of Ni, Al, Cu, using controlled pulsed laser exposure and high-resolution atomic force microscopy (AFM). The resulting framework connects thin-film microstructural evolution to broader aerospace-relevant phenomena.

ENGINEERING PORTFOLIO

<https://joseosa.github.io/Engineering-Portfolio/index.html>

LANGUAGES

Fluent in English | Fluent in Spanish | German | French